磁和聚合

Cloud Manager 3.7

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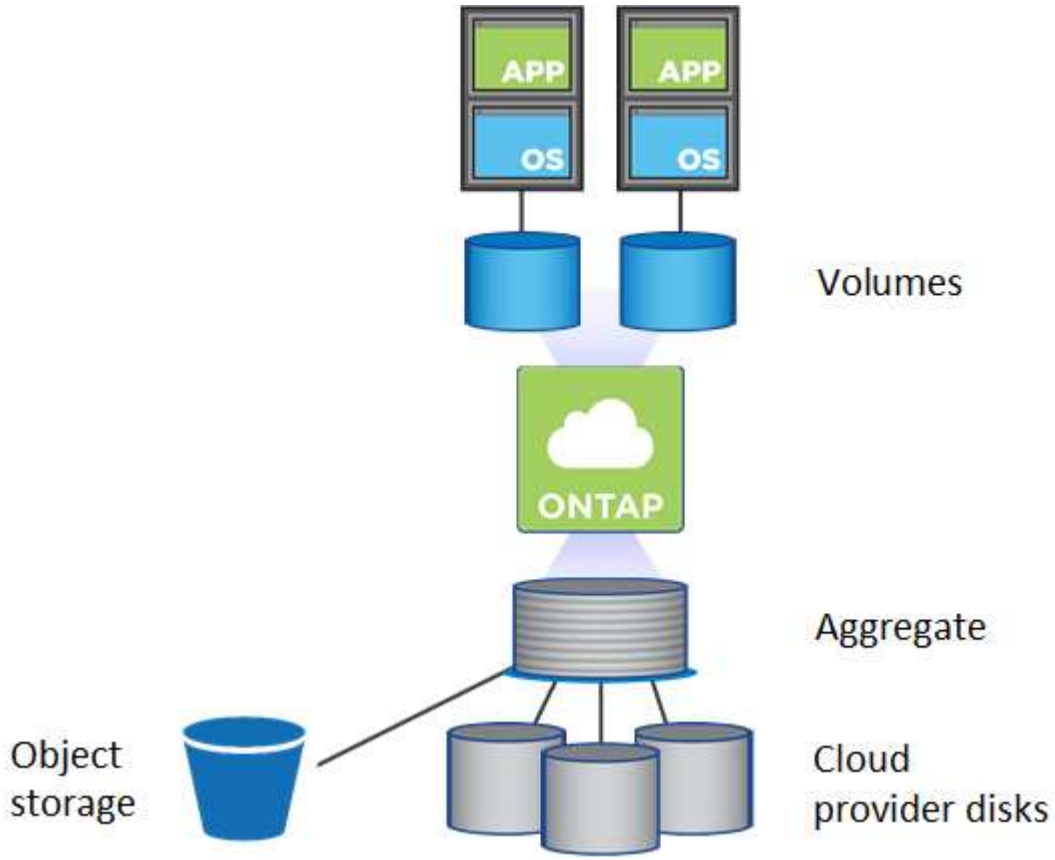
磁□和聚合

了解 Cloud Volumes ONTAP 如何使用云存□可以□助□了解存□成本。

概述

Cloud Volumes ONTAP 使用云提供商存□作□磁□，并将其分□□一个或多个聚合。聚合可□一个或多个卷提供存□。



支持多□□型的云磁□。□可以在□建卷□□□磁□□型，在部署 Cloud Volumes ONTAP □□□默□磁□大小。

> 从云提供商□□的存□□量□ *raw capacity* 。可用容量 _ □小，因□□留□ Cloud Volumes ONTAP 使用的□□□□ 12 到 14% 。例如，如果 Cloud Manager □建了一个 500 GB 聚合、□可用容量□ 442.94 GB 。

AWS 存□

在 AWS 中， Cloud Volumes ONTAP 将 EBS 存□用于用□数据，并将本地 NVMe 存□用作某些 EC2 □例□型的 Flash Cache 。

EBS 存□

底□ EBS 磁□□型可以是通用 SSD 、配置的 IOPS SSD 、吞吐量□化 HDD 或冷 HDD 。□可以将 EBS 磁□与 Amazon S3 配□ "将非活□数据分□到低成本□象存□"。

在□高□□上、 EBS 磁□□型之□的区□如下所示：

- 通用 SSD_ 磁□可平衡各□工作□□的成本和性能。性能是根据 IOPS 来定□的。
- *provisioned IOPS SSD* 磁□□用于需要以□高成本□得最高性能的□□□用程序。
- *Throughput Optimized HDD* 磁□□用于□常□□的工作□□，□些工作□□需要以更低的价格□□快速一致的吞吐量。
- 冷 HDD_ 磁□用于□□或不常□□的数据，因□性能非常低。与吞吐量□化的 HDD 磁□一□、性能也是根据吞吐量来定□的。

> HA 配置和数据分□不支持冷 HDD 磁□。

本地 **NVMe** 存□

某些 EC2 □例□型包括本地 NVMe 存□， Cloud Volumes ONTAP 将其用作 "Flash Cache"。

- 相□□接 *
- "AWS 文□： EBS 卷□型"
- "了解如何在 AWS 中□□的系□□□磁□□型和磁□大小"
- "□看 AWS 中 Cloud Volumes ONTAP 的存□限制"
- "□看 AWS 中支持的 Cloud Volumes ONTAP 配置"

Azure 存□

在 Azure 中，一个聚合最多可以包含 12 个大小相同的磁□。磁□□型和最大磁□大小取决于□使用的是□□点系□□是 HA □：

□□点系□

- _Premium SSD 受管磁□ _ 以□高的成本□ I/O 密集型工作□□提供高性能。
- □准 SSD 受管磁□ _ 可□需要低 IOPS 的工作□□提供□定一致的性能。
- 如果□不需要高 IOPS 并希望降低成本， _Standard HDD 受管磁□ _ 是一个不□的□□。

  □个受管磁□□型的最大磁□大小□ 32 TB 。

  □可以将受管磁□与 Azure Blob 存□配□ "将非活□数据分□到低成本□象存□"。

HA □

HA □使用高□□面 Blobs， □些□面的最大磁□大小□ 8 TB 。

- 相□□接 *
- "Microsoft Azure 文□： Microsoft Azure 存□□介"
- "了解如何在 Azure 中□□的系□□□磁□□型和磁□大小"
- "□看 Azure 中 Cloud Volumes ONTAP 的存□限制"

GCP 存□

在 GCP 中，一个聚合最多可以包含 6 个大小相同的磁□。最大磁□大小□ 16 TB 。

磁□□型可以是 *zonal SSD persistent disks_or _zonal standard persistent disks*。□可以将永久性磁□与 Google 存□分段配□ "将非活□数据分□到低成本□象存□"。

- 相□□接 *
- "Google Cloud Platform 文□：存□□□"
- "□看 GCP 中 Cloud Volumes ONTAP 的存□限制"

RAID □型

□个 Cloud Volumes ONTAP 聚合的 RAID □型□ RAID0 （条□化）。不支持其他 RAID □型。Cloud Volumes ONTAP 依□云提供商提供磁□可用性和持久性。

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